

Analysis of Per- and Polyfluoroalkyl Substances (PFAS) in Accordance With EPA 1633 Part 2

Analysis of Aqueous Matrices using the Waters Xevo™ TQ MS

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US EPA Method 1633 has become the foundation method for analysis of PFAS in non-potable water matrices, soils, biosolids and tissues in the United States. The method consists of sample preparation using weak anion exchange (WAX) solid phase extraction (SPE) with graphitized carbon black (GCB) clean up. This application note is the second in a series demonstrating a comprehensive solution for performing the EPA 1633 methodology. The focus of this note is preparation and analysis of authentic water samples utilizing a prototype bilayer dual-phase SPE cartridge and LC-MS/MS method on an ACQUITY™ Premier BSM FTN LC System coupled to a Xevo™ TQ Absolute Tandem Quadrupole Mass Spectrometer.

Keywords: PFAS (Per- and Polyfluoroalkyl Substances), EPA Method 1633, LC-MS/MS (Liquid Chromatography-Tandem Mass Spectrometry), Solid phase extraction (SPE), wastewater analysis, Waters

Introduction

US EPA Method 1633 was first introduced in August 2021 to become the foundational method for analysis of PFAS in non-potable water matrices, soils, biosolids, and tissues.¹ At the time of writing this document, Method 1633 is in the 4th Draft Phase with the final version expected to be released at the end of 2023.² By the final release of EPA 1633, it will have been multi-lab validated for each type of sample matrix included in the method. The method covers 40 PFAS and utilizes isotope dilution calibration and quantitation. Required sample preparation differs slightly depending on sample type

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but all sample types utilize SPE on a WAX cartridge in combination with GCB clean up. EPA 1633 was created to support sample analysis for the Clean Water Act (CWA) and Department of Defense (DoD) monitoring and remediation, but it covers such a wide range of matrices and compounds that its applicability is expected to be widespread.

This is the second in a series of application notes addressing sample preparation, analysis and method performance of EPA 1633 using a comprehensive workflow of Waters technologies. This application note will focus on the preparation of authentic water samples with analysis utilizing the LC-MS/MS method established in Part 1 on an ACQUITY Premier BSM FTM UPLC System coupled to a Xevo TQ Absolute Mass Spectrometer.³ The use of a combined WAX and GCB sample extraction and clean-up workflow is demonstrated on ground water, surface water, and wastewater (influent and effluent).

Experimental

Sample Preparation

Samples discussed in this application note include ground water and surface water that were collected locally, and influent and effluent wastewater that were kindly provided by a municipal wastewater treatment facility in the Midwest United States. All water samples were collected using grab sampling directly into 250 mL high density polypropylene bottles. Samples were frozen until sample analysis according to EPA 1633 guidelines and holding times. Sample bottles were weighed prior to sample preparation (full) and after sample preparation (empty) to determine the exact volume collected in each bottle. In addition to authentic samples, the Waters ERA PFAS in Wastewater certified reference material (CRM) was processed with the samples.

A prototype bilayer dual-phase SPE cartridge containing both Weak Anion Exchange (WAX) and Graphitized Carbon Black (GCB) sorbents was used for sample preparation instead of adding dispersive GCB into the

sample for cleanup. Full sample preparation details are listed in Figure 1 and are adapted directly from the EPA 1633 method. Two changes to the method were made including a sample volume change and combining the dispersive GCB step into the SPE cartridge, as described previously. Sample volume was reduced to 250 mL from the suggested 500 mL sample in the method due to instrument sensitivity of the Xevo TQ Absolute Mass Spectrometer. Reducing sample volume reduces sample shipping and storage costs, as well as allowing for faster sample preparation when loading half the volume of sample onto an SPE cartridge. Combining the GCB and WAX into the same cartridge provides the convenience of minimizing complications from using loose material and reducing the number of steps during sample preparation without compromising the method.

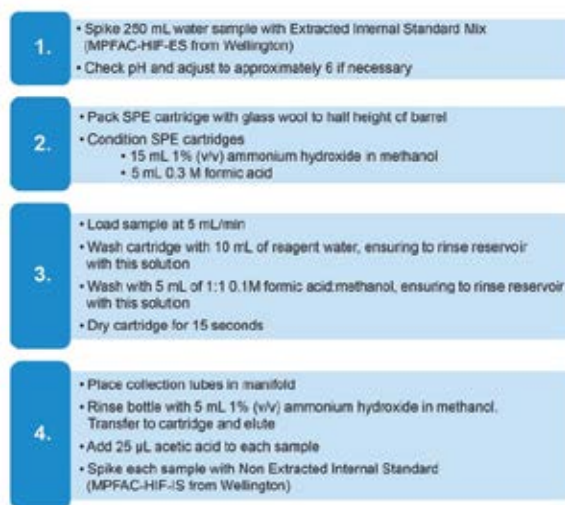


Figure 1. Full method details of the sample preparation process used for all water samples. Adapted from EPA Method 1633.

All samples were spiked with 5 ng/L (sample concentration equivalent) of the required extracted internal standard (EIS) prior to extraction and 5 ng/L (sample concentration equivalent) of the required non-extracted internal standard (NIS) after extraction. The calibration curve range for each analyte is listed in Appendix Table 2 and was determined from the data acquired and presented in Part 1 of this application note series.³ All standards were obtained as mixes from Wellington Laboratories.

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LC Conditions

LC system	ACQUITY Premier BSM with FTN
Vials	700 µL Polypropylene Screw Cap Vials
Analytical column	ACQUITY Premier BEH™ C18 2.1 x 50 mm, 1.7 µm
Isolator column	Atlantis™ Premier BEH C18 AX 2.1 x 50 mm, 5.0 µm
Column temperature	35 °C
Sample temperature	10 °C
PFAS kit	PFAS Install Kit with OASIS WAX 150 mg
Injection volume	2 µL
Flow rate	0.3 mL/min
Mobile phase A	2 mM ammonium acetate in water
Mobile phase B	2 mM ammonium acetate in acetonitrile

Gradient Table

Time (min)	%A	%B	Curve
0	95	5	initial
0.5	75	25	6
3	50	50	6
6.5	15	85	6
7	5	95	6
8.5	5	95	6
9	95	5	6
11	95	5	6

MS Conditions

MS system	Xevo TQAbsolute
Ionization mode	ESICapillary
Voltage	0.5 kV
Source temperature	100 °C
Desolvation temperature	350 °C
Desolvation flow	900 L/hr
Cone flow	150 L/hr
MRM method	See Appendix for Full MRM Method details

Data Management

Software	waters_connect™ for Quantitation
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Results and Discussion

Recovery in Water Samples

EPA 1633 is a performance-based method that allows modifications as long as the performance criteria outlined in the method are all met. One major modification presented in this work is to use a bilayer dual-phase SPE cartridge that combines the otherwise dispersive GCB clean up step into the WAX SPE cartridge. GCB is difficult to work with and accurately measure, therefore utilizing a bilayer cartridge eliminates the untidy dispersive step.

More importantly, combining the GCB cleanup step into the SPE extraction saves valuable time in the laboratory during the sample preparation process. Additionally, less preparation steps allow for fewer opportunities for introduction of unintended PFAS sample contamination.

One of the important performance criteria that must be established in order to prove equivalence of this approach is the extracted internal standard (EIS) and non-extracted internal standard (NIS) recovery acceptance limits in the 4th Draft Method 1633 (See Table 6 within that document).¹ The individual recovery performance of the bilayer dual-phase SPE cartridge for ground water, surface water (with high organic matter content), influent wastewater (settled only), and effluent wastewater (fully treated discharge water) are listed for each EIS and NIS in Table 1.

The data reported in Table 1 is the average recovery and %RSD for five replicate extractions of each matrix type. The mean recovery of all EIS among the 20 samples extracted was 91.2% with a mean RSD of 9.2%.

Compound	Ground water		Surface water		Influent water		Effluent water	
	Average recovery (%)	%RSD	Average recovery (%)	%RSD	Average recovery (%)	%RSD	Average recovery (%)	%RSD
¹² C ₈ -PFBA	100.0	2.7	111.9	8.1	85.8	9.2	86.6	13.8
¹² C ₈ -PFPeA	98.6	4.2	110.1	8.5	101.6	5.7	100.3	15.1
¹² C ₈ -PFHxA	97.2	3.1	111.2	8.0	111.0	8.4	102.5	14.3
¹² C ₈ -PFHpA	97.3	4.5	108.8	9.2	111.0	8.5	99.7	14.8
¹² C ₈ -PFOA	98.8	2.5	110.6	9.0	113.0	13.3	100.7	16.5
¹² C ₈ -PFNA	96.6	5.2	110.2	11.3	112.0	18.0	101.1	12.5
¹² C ₈ -PFDA	92.1	2.9	108.1	9.5	103.8	18.1	97.0	17.5
¹² C ₈ -PFUnDA	88.5	2.9	102.0	4.8	93.0	18.7	91.6	15.0
¹² C ₈ -PFDoDA	83.1	2.8	89.8	8.5	83.8	18.2	82.5	14.0
¹² C ₈ -PFTeDA	72.5	5.0	66.7	10.1	32.3	17.6	52.5	13.4
¹² C ₈ -PFBS	97.8	2.3	110.9	6.5	116.8	12.3	102.6	13.5
¹² C ₈ -PFHxS	97.0	5.0	113.1	6.9	112.5	8.3	104.8	14.5
¹² C ₈ -PFOS	93.2	1.8	108.7	7.8	108.5	13.9	97.8	17.3
¹² C ₈ -4:2 FTS	82.8	6.9	92.1	5.1	179.8	9.1	102.8	21.4
¹² C ₈ -6:2 FTS	94.1	2.8	95.0	5.0	197.8	11.0	101.9	19.1
¹² C ₈ -8:2 FTS	91.5	4.2	91.8	7.5	149.7	16.6	90.4	12.9
¹² C ₈ -FOSA	92.3	3.6	99.4	5.8	101.3	19.8	96.7	18.7
¹² C ₈ -GenX	98.0	3.5	105.9	6.6	85.2	7.5	98.6	14.8
D ₈ -N-EtFOSAA	91.3	3.4	91.9	6.0	127.0	15.4	93.3	18.3
D ₈ -N-MeFOSAA	89.0	4.8	90.3	5.5	137.4	16.2	91.6	20.3
d ₈ -NMeFOSA	63.5	5.9	64.0	13.2	52.8	24.3	82.3	19.2
d ₈ -NEtFOSA	61.1	7.0	61.9	13.5	38.9	23.8	78.5	19.6
d ₇ -NMeFOSE	70.0	5.7	74.0	11.8	56.9	20.3	74.5	17.7
d ₉ -NEtFOSE	67.2	5.9	71.1	12.3	52.9	20.2	71.1	18.3
¹² C ₈ -PFBA	134.8	20.0	112.2	6.5	89.6	6.6	114.1	7.1
¹² C ₈ -PFHxA	133.9	19.4	117.4	5.7	120.8	7.2	124.7	5.8
¹² C ₈ -PFOA	127.9	22.7	116.3	7.8	132.2	8.1	123.5	8.0
¹² C ₈ -PFNA	133.3	21.3	115.1	9.1	132.8	4.5	122.3	7.5
¹² C ₈ -PFDA	136.8	20.3	115.8	6.3	155.9	4.0	124.4	7.7
¹⁸ O ₂ -PFHxS	133.6	21.4	115.7	7.4	112.2	7.0	119.8	6.6
¹² C ₈ -PFOS	132.2	22.7	116.1	6.6	118.9	7.2	121.6	8.2

Table 1. Average recovery of the extracted internal standards (EIS) and non-extracted internal standards (NIS) using the bilayer dual-phase SPE cartridge for each water sample type evaluated (n=5).

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Figure 2 directly compares the average recovery across all water sample types with the allowable recoveries in EPA 1633 Table 6 (Draft 4). The recoveries in water samples were easily within the recovery acceptance limits for each compound, and in all cases were significantly above the minimum recovery level. This demonstrates that even in the more complex water matrices, the cartridge has equivalent performance as using dispersive GCB and is fit-for-purpose.

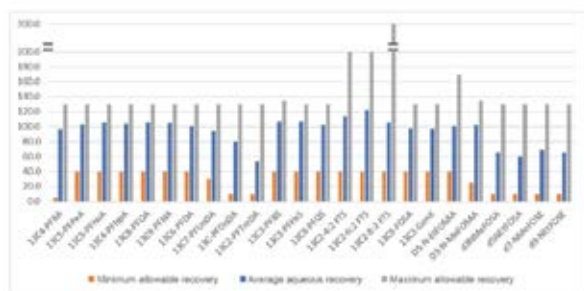


Figure 2. Average recovery of the extracted internal standards (EIS) in all four aqueous sample types (blue) compared to the minimum (orange) and maximum (gray) percent recoveries allowed in the EPA 1633 method (n=20). Note the split axis to accommodate the maximum recovery value for 13C2-8:2 FTS.

Analysis of a Certified Reference Material

Accuracy of analysis is important for quantitating PFAS in customer samples. A certified reference material (CRM) from Waters ERA was processed with the authentic samples as a benchmark for workflow accuracy. The PFAS in Wastewater CRM is certified for all EPA 1633 analytes, giving a representative reference material for method performance without having to spike unknown matrix samples which can become complicated without a sample free from PFAS.

Figure 3 shows the average quantitative results for three replicate extraction and analyses of the Wastewater CRM. The dotted and dashed red lines indicate the minimum and maximum certified value range of the CRM. The solid blue line represents the certified value. The solid gray line represents the average experimental quantitated value determined during sample analysis. All 40 target PFAS in EPA 1633 were quantified within the allowable minimum and maximum concentration

range with a mean trueness of 92% and trueness range of 73–112%. This demonstrates confidence in accuracy of the sample preparation, analysis and data processing workflow using Waters solutions.

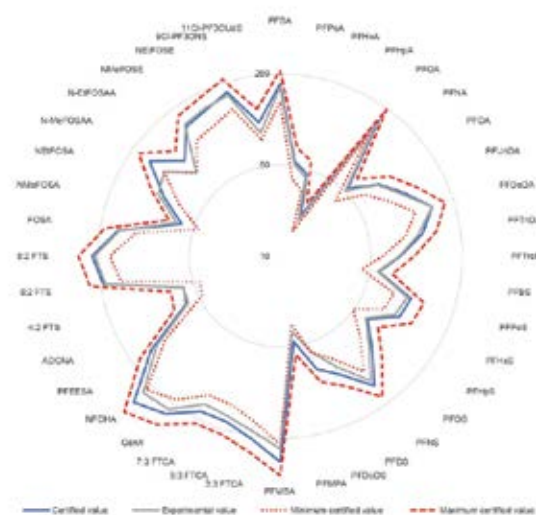


Figure 3. Quantified values of all 40 EPA 1633 target analytes in Waters ERA PFAS in Wastewater CRM. Red lines represent the minimum (dotted) and maximum (dash) certified value range of the CRM. The blue line represents the certified value. The solid gray line represents the average experimental quantitated value (n=3). Note the axis is represented using a log scale.

Analysis of PFAS in Authentic Water Samples

The presented workflow was utilized to detect and quantify the 40 target PFAS analytes in ground water, surface water, influent water, and effluent water.

The LC gradient method was designed to provide a minimum of 1 minute separation between potential cholic acid interferences and PFOS.² Because cholic acids are produced to aid digestion, they can be present in large amounts in wastewater samples.

Figure 3 demonstrates the presence of cholic acids in the influent and effluent wastewater samples and the successful separation of them from interference with PFOS. PFAS were detected in all samples, as highlighted in Table 2.

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Analyte name	Ground water		Surface water		Influent wastewater		Effluent wastewater	
	Concentration (ng/L)	%RSD	Concentration (ng/L)	%RSD	Concentration (ng/L)	%RSD	Concentration (ng/L)	%RSD
PFBA	7.03	3.0	15.36	8.4	21.93	16.3	21.15	7.7
PFPeA	2.47	2.5	4.30	1.7	11.55	15.5	12.12	3.7
PFFrA	1.67	3.0	3.53	2.7	15.98	17.4	23.81	1.7
PfHpA	1.01	8.0	2.32	5.0	3.33	13.4	3.33	1.6
PFOA	0.69	10.0	4.98	6.7	27.02	14.5	14.89	6.5
PFNA	N.D.	-	0.77	15.0	0.91	11.4	1.28	20.2
PFDA	N.D.	-	0.51	32.1	0.95	4.5	2.44	20.7
PFluDA	N.D.	-	N.D.	-	N.D.	-	N.D.	-
PFDODA	N.D.	-	N.D.	-	BLoQ	-	N.D.	-
PFTriDA	N.D.	-	N.D.	-	N.D.	-	N.D.	-
PFTreDA	N.D.	-	N.D.	-	N.D.	-	N.D.	-
PFBS	1.05	7.1	2.93	6.0	41.12	12.6	36.16	3.9
PFPeS	0.11	8.7	0.17	11.8	0.39	23.7	0.34	6.2
PFFrS	0.26	10.3	0.86	5.6	4.72	17.7	4.07	3.5
PfHpS	N.D.	-	N.D.	-	N.D.	-	BLoQ	-
PFOs	0.34	16.4	3.40	22.4	3.17	9.4	6.17	19.9
PFNS	N.D.	-	N.D.	-	N.D.	-	N.D.	-
PFDs	N.D.	-	N.D.	-	N.D.	-	N.D.	-
PFDoS	N.D.	-	N.D.	-	N.D.	-	N.D.	-
GenX	N.D.	-	0.32	8.8	0.65	15.2	0.59	5.4
ADONA	N.D.	-	N.D.	-	N.D.	-	N.D.	-
9ClPF3ONS	N.D.	-	N.D.	-	N.D.	-	N.D.	-
11ClPF3OUs	N.D.	-	N.D.	-	N.D.	-	N.D.	-
4_2 FTS	N.D.	-	N.D.	-	N.D.	-	N.D.	-
8_2 FTS	N.D.	-	N.D.	-	4.44	16.2	2.91	18.9
8_2 FTS	N.D.	-	N.D.	-	N.D.	-	N.D.	-
FOSA	BLoQ	-	BLoQ	-	BLoQ	-	BLoQ	-
NMeFOSA	N.D.	-	N.D.	-	N.D.	-	N.D.	-
NEtFOSA	N.D.	-	N.D.	-	N.D.	-	N.D.	-
NMeFOSAA	N.D.	-	N.D.	-	1.16	10.7	1.26	28.1
NEtFOSAA	N.D.	-	N.D.	-	0.90	14.6	0.90	27.6
NMeFOSE	N.D.	-	N.D.	-	2.96	9.3	BLoQ	-
NEtFOSE	N.D.	-	N.D.	-	N.D.	-	N.D.	-
3_3 FTCA	N.D.	-	N.D.	-	4.91	21.4	N.D.	-
5_3 FTCA	N.D.	-	N.D.	-	88.91	18.4	3.91	4.3
7_3 FTCA	N.D.	-	N.D.	-	4.66	16.8	N.D.	-
PFMPA	N.D.	-	N.D.	-	N.D.	-	N.D.	-
PFMBA	N.D.	-	N.D.	-	N.D.	-	N.D.	-
PFEESA	N.D.	-	N.D.	-	N.D.	-	N.D.	-
NFDHA	N.D.	-	N.D.	-	N.D.	-	N.D.	-

Table 2. Detected concentrations of PFAS in each type of water sample and associated %RSD of n=5 replicates. (N.D.) not detected. (BLoQ) below limit of quantitation.

Each sample was collected and extracted in five replicates and the average calculated concentration is reported with associated %RSDs for the replicates.

The ground water had the lowest detectable PFAS with nine PFAS detected above the limit of quantitation (LOQ) and ranging from 0.11–7.03 ng/L.

The surface water sample had a slightly larger range of PFAS, with 12 detected above the LOQ in a range of 0.17–15.4 ng/L. The wastewater samples had the largest range and concentrations of PFAS detected. A comparison of the detected PFAS in the influent and effluent water is shown in Figure 5.

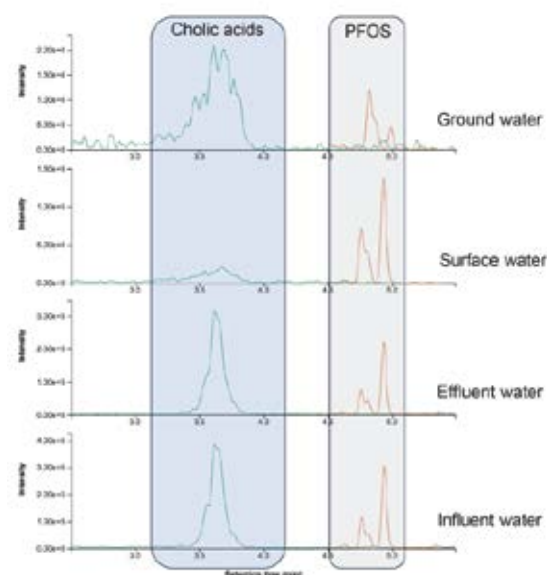


Figure 4. Overlay chromatograms of cholic acids and PFOS MRM channels demonstrating the large cholic acid interference present in wastewater samples.

